NIHR | National Institute for Health and Care Research

Summary

Reference Number	NIHR306500
Lead Applicant	Dr Chris Carvalho
Research Title	Real-World Health and Financial Impact of Medication Optimisation on kidney and cardiovascular disease: A Comprehensive Analysis in an Ethnically Diverse UK Urban Population
Plain English Summary	**Background** Chronic Kidney Disease is a serious health issue that can lead to kidney failure and heart problems if not properly treated. If the kidneys stop working, the person will need to have treatment to replace the kidney function, either by a kidney transplant or dialysis. As well as kidney failure, people with chronic kidney disease are at greater risk of heart attacks, heart failure, strokes and damage to the blood supply in their legs and feet. Some people from different backgrounds are affected more. Older people, men and people from Black or South Asian backgrounds have a greater risk of kidney failure and heart disease. A range of medicines are used by General Practitioners to prevent people from having worsening kidney disease or developing heart disease. They improve blood pressure control, lower how much protein is in the urine, improve how well the kidney filters the blood, lower cholesterol, and lower sugar in the blood. Some of the medications are new and the impact of combinations of these medicines is unclear. More research is needed to find out whether some groups of people are undertreated and the impact that best treatment could have on reducing inequalities, costs and hospital use. This is important as it would help doctors and patients make better decisions about which groups are the most undertreated and which medications to use together to best reduce the risk of disease based on personal characteristics such as age, sex and ethnic background. **Aim of the research** This study aims to investigate how age, sex, ethnicity, and socioeconomic factors influence the effects of medications on the kidneys and heart in individuals with chronic kidney disease. It will evaluate impact of key medications used alone or in combination. Additionally, the study seeks to estimate the health and financial implications of using these medications optimally in a diverse urban population like North East London. The research will provide insights that can contribute to more personalized and effective treatment strategies for patients, taking into account the diverse characteristics of the population. **Design and Methods** I will review the latest evidence to find out what we know about how these medications work in people from different backgrounds and when used together. I will review the medical records of people with chronic kidney disease in North East London to find out if there are differences in how the key medications are provided to people with different backgrounds including on age, sex, ethnicity, or socioeconomic factors. I will then estimate what the likely kidney and heart benefits would be if we were to give the medications to everyone who should be on them and calculate the costs of doing so, which includes the savings from reduced hospital care. **Patient and Public Involvement** This summary was written with a group of people with lived experience of kidney and heart disease who believe this is an important research topic, and who I consulted with in the design of this research. I will continue to work with this group of expert patient advisors to use their knowledge and experience of accessing health care to help shape and guide the research and develop accessible summaries of the results. **Dissemination** I will share this information via published papers, in conference presentations and at patient groups guided by the expert patient advisors. This knowledge will help doctors tailor treatments and give patients the right information about what works for them, to support shared decision making, improving their quality of life and reducing the differences some receive in health outcomes.

Our purpose for collecting this data is to communicate with you about your application and have the necessary information to evaluate you for a grant. The data we collect here is collected in the public interest. Information provided here may be subject to Freedom of Information requests.

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Table Of Contents

1. Application Summary Information
2. Applicant CV
3. Applicant Research Background
4. Plain English Summary of Research
5. Scientific Abstract
6. Detailed Research Plan
7. Patient & Public Involvement
8. Training & Development and Research Support
9. Detailed Budget
10. Uploads
11. Participants and Signatory
12. Acknowledgement, Review and Submit

1. Application Summary Information

Host organisation (which will administer any award)
Queen Mary University of London

Partner organisation (if applicable)	

Application Type

Strategic Theme	Health data science

Partnership Fellowship	N/A

WTE % Please select which WTE option you would prefer
0.80

Duration Doctoral Fellowship is a 3 year award and based on your WTE preference the duration in months is calculated as follows
45

Research Title	Real-World Health and Financial Impact of Medication Optimisation on kidney and cardiovascular disease: A Comprehensive Analysis in an Ethnically Diverse UK Urban Population

Research Type	Secondary data analysis, Cohort study, Other (please specify)

Please provide more details on your selection above.
and Economic Analysis

Proposed start date if grant awarded	01/10/2025

2. Applicant CV

A CV for the Applicant is a mandatory component of this application; please ensure this is completed fully.
Please note that some of the CV questions are automatically completed from information you have provided in the 'Update CV' section of the 'Manage My Details' page. These can be updated via the ARAMIS Home Page or by following this link https://aramis.nihr.ac.uk/MyAccount/UpdateCV.aspx

The 'Publication Record' section of the form is automatically populated from publications added into the "My Research Outputs" page of your ARAMIS account.

Degrees and Professional Qualifications

From	To	Qualification	Subject	Country	Institution	Class	Department
09/2014	08/2019	MSc	Public Health	United Kingdom (England)	Univeristy of London/LSHTM	Pass	Public Health
09/2006	07/2007	Diploma	Obstetrics and Gynaecology	United Kingdom (England)	RCOG	Pass	RCOG
10/2005	08/2008	MRCGP	General Practice	United Kingdom (England)	RCGP	Pass	General Practice
09/1999	08/2004	MBChB	Medicine	United Kingdom (England)	The University of Birmingham	Pass	School of Medicine

Present and previous positions
(including from/to dates, job title and employer)

From	To	Position	Organisation	WTE % Research
01/2022	-	Cardiovascular and Renal Clinical Lead	North East London ICB	0
10/2021	10/2023	Research Fellow	Queen Mary University, London	50
01/2021	03/2023	Primary Care CVD Lead and BPM@H London Clinical Reference Group Chair	NHS England	0
05/2019	-	GP	De Beauvoir Surgery	0
05/2019	-	Clinical Lead and Research Fellow	Queen Mary University, London	25
09/2018	12/2021	Senior Clinical Advisor	NHS England	0
04/2017	04/2018	GP Quality Improvement Lead	UCL Partners	0
09/2014	-	Planned Care Lead and LTC Lead	City and Hackney Place based Partnership, NEL ICB	0
09/2014	09/2015	Specialist Advisor	Care Quality Commission	0
09/2014	12/2015	GP	City and Hackney GP Confederation	0
04/2013	04/2014	GP	Midway Surgery	0
07/2012	06/2013	North Locality Lead for the Integrated Care	Islington CCG	0
11/2011	09/2015	Chief Medical Officer	Spinach Health	0
11/2008	04/2013	GP	Hanley Primary Care Centre	0
10/2005	10/2008	GP Vocational Training Scheme	NWLH NHS Trust	0
08/2004	08/2005	House Officer	Shrewsbury and Telford	0

Present and previous positions

(including from/to dates, job title and employer)

From	To	Position	Organisation	WTE % Research
			Hospital NHS Trust	

Research Grants held

Title of Grant	Role in the Research Grant	Total Grant £	Source of Grant (Organisation Name)	Duration (In Months)	Start Date of Funding	End Date of Funding
SPCR GP Career Progression Award - Improving Primary Care Case-Finding for Familial Hypercholesterolaemia	Primary Investigator	144300	NIHR	24	01/09/2021	01/09/2023

Publication Record

(include the list of authors, article title, journal name, and volume)
This information can be edited via the “My Research Outputs” page of your ARAMIS account.

Publications
Rison, SC., Mathur, R., Carvalho, C. and Robson, JP.. (2023) ‘Inequities in hypertension: we can do better ... but how?’, *The British journal of general practice : the journal of the Royal College of General Practitioners*, 73(736).
Rison, S., Redfern, O., Dostal, I., Carvalho, C., Mathur, R., Raisi-Estabragh, Z. and Robson, J.. (2023) ‘Inequities in hypertension management: observational cross-sectional study in North East London using electronic health records.’, *The British journal of general practice : the journal of the Royal College of General Practitioners*, 73(736).
Roddick, AJ., Wonnacott, A., Webb, D., Watt, A., Watson, MA., Staplin, N., Riding, A., Lioudaki, E., Kuverji, A., Kossi, ME., Holmes, P., Holloway, M., Fraser, D., Carvalho, C., Burton, JO., Bhandari, S., Herrington, WG. and Frankel, AH.. (2023) ‘UK Kidney Association Clinical Practice Guideline: Sodium-Glucose Co-transporter-2 (SGLT-2) Inhibition in Adults with Kidney Disease 2023 UPDATE.’, *BMC nephrology*, 24(1).
Rison, SC., Redfern, O., Mathur, R., Dostal, I., Carvalho, C., Raisi-Estabragh, Z. and Robson, JP.. (2023) ‘COVID pandemic impact on hypertension management in North-East London: an observational cohort study using electronic health records’,
Xie, CX., Robson, J., Williams, C., Carvalho, C., Rison, S. and Raisi-Estabragh, Z.. (2022) ‘Dual antithrombotic therapy and gastroprotection in atrial fibrillation: an observational primary care study.’, *BJGP Open*, 6(4).
Rison, SC., Carvalho, C., Rull, G. and Robson, J.. (2022) ‘Investigating hypertension in younger patients.’, *BMJ (Clinical research ed.)*, 376.
Wu, R., Rison, SCG., Raisi-Estabragh, Z., Dostal, I., Carvalho, C., Robson, J. and Mihaylova, B.. (2021) ‘Gaps in antihypertensive and statin treatments and benefits of optimisation: a modelling study in a 1 million ethnically diverse urban population in UK.’, *BMJ open*, 11(12).
Carvalho, C., Williams, C., Raisi-Estabragh, Z., Rison, S., Patel, RS., Timmis, A. and Robson, J.. (2021) ‘Application of a risk stratification tool for familial hypercholesterolaemia in primary care: an observational cross-sectional study in an unselected urban population.’, *Heart (British Cardiac Society)*, 107(15).
Hull, SA., Williams, C., Ashworth, M., Carvalho, C. and Boomla, K.. (2020) ‘Prevalence of suspected COVID-19 infection in patients from ethnic minority populations: a cross-sectional study in primary care.’, *The British journal of general practice : the journal of the Royal College of General Practitioners*, 70(699).

Relevant Prizes, Awards and other Academic Distinctions

Please note the information for additional prizes or awards is automatically populated from information provided in the "Update CV" section of the "Manage My Details" page, which you can update by following this link https://aramis.nihr.ac.uk/MyAccount/UpdateCV.aspx

Date (mm/yyyy)	Source	Prize	Role of Applicant

ORCID iD 0000-0002-4114-9210

3. Applicant Research Background

Applicant Professional Background	Medically Qualified

Specialty (if Medically Qualified)
General Practice

Do you have a GMC number?	Yes

GMC number Please supply a valid GMC number	6106295

Please describe your research career to date

Since 2019 I have been a Research Fellow and Clinical Lead in the Clinical Effectiveness Group (CEG), in Queen Mary University, London (QMUL). CEG uses primary care health data for research, focusing on reducing health inequalities and improving population health for over two million people in North East London. I have been working with academics, clinicians, health data scientists, analysts and facilitators to turn knowledge and data into clinical improvements. I have worked with colleagues on observational studies using healthcare data and have helped develop research projects.

I have personal and clinical experience of ethnically related deprivation in relation to disease and in particular cardiovascular disease and chronic kidney disease. My clinical training has shown the ability to make improvements in outcomes, but also the need to think differently and make treatments accessible to people from different backgrounds. I have dedicated my clinical and research time to developing interventions and uncovering evidence to address inequalities.

I have built on the foundation of my Master's in Public Health which included population health management, economic evaluation, statistical analysis, and health policy. In assessing the prevalence of COVID-19, with my co-authors, I used logistic regression and multivariate analysis, and my primary care experience of coding to analyse and interpret the rates of respiratory tract symptoms and infections early in the pandemic.(1) We found that patients of South Asian and Black ethnicity were at increased risk of a clinical diagnosis of COVID-19 in primary care, independent of other factors.

In 2020-21 I led research on case-finding of familial hypercholesterolaemia from design to publication. In the real-world application of a risk prediction algorithm for familial hypercholesterolaemia identification, I gained experience in handling large scale real-world 'big data' sets from primary care health records, including data accuracy and imputation methods for missing data.(2) Recently, I was co-author on a paper on inequities in hypertension using primary care data. I analysed the data on management and assessed variation by ethnic group, finding that although those from Black backgrounds had the same level of treatment intensity, their blood pressure control was worse compared to those from other ethnic groups.(3) . In both, I was able to use my data analysis skills and advanced them further, gained experience in leading a research project, and linked my clinical and research roles.

I was involved in developing a clinical guideline on Sodium-Glucose Co-transporter-2 Inhibition in Adults with Kidney Disease.(4) I reviewed and synthesised the evidence on complications and analysed data from trials on incidence and risk. Then I made recommendations of how to consider the risks and mitigate their occurrence. This

gave me experience of evidence synthesis in assessing quality and relevance of studies, summarizing and interpreting results. I also contributed to an NIHR and Health Data Research UK paper on use of primary care databases for research.(5). I gained experience in both of these papers of working as a research collaborator, learning a lot from those involved and gaining different perspectives.

I secured an NIHR School of Primary Care Research Career Progression Fellowship to study familial hypercholesterolaemia risk prediction using primary care data, to prepare me for a PhD, and pursuing an academic career. I reviewed the current evidence before designing and implementing an online questionnaire, enabling patients to input their family histories. Using a before and after study design, I evaluated the completion of the questionnaire by age, sex, ethnicity and co-morbidities. I found that overall, a small number of patients used the questionnaire without prompting but the coding guidance aided the correct history recording. I was able to instigate a change to a national indicator on FH, allowing primary care to assess risk rather than having to refer to secondary care, thereby contributing to better population health outcomes and reduce burden on hospitals.(6)

I have mentored GPs working on quality improvement projects or research topics, creating an educational programme and a supervision structure to develop them and ensure their success. Plans are underway to recruit more GPs for the programme in the coming year. I have also developed educational meetings for GPs and for patients and have led the development of end-to-end clinical pathways to ensure optimal clinical management.

I am an experienced frontline GP and Associate Clinical Director for North East London Integrated Care Board (ICB) leading on initiatives with clinicians and patients, to reduce renal and cardiovascular disease by focusing on prevention, innovative ideas and translating research into actionable clinical improvement. I have been developing leadership, policy and research skills and applied these to real world health care, in the implementation and evaluation of new interventions in diverse population contexts which reduce health inequalities, including, a £300,000 initiative to improve anticoagulation, and a £2 million renal programme developed across London. I have used population health data analysis to identify areas of need, share learning and driven improvement with a learning health system approach to improve care and reduce unwarranted variation. I contributed significantly to our region's exceptional post pandemic recovery; our ICB was ranked by CVDPrevent as the top for almost all major cardiovascular, blood pressure and lipid lowering attainment measures.

The fellowship will help to build on the translational research skills and leadership experience gained from my previous work and provide me with new skills and expertise in health economic impact modelling and statistical analysis. Utilizing a range of supporting evidence, I aim to turn this knowledge into successfully influencing policy and guidelines and support my development as an independent researcher and leader in primary care research and health policy.

Have you already registered for a PhD?	No

Has this application, or a similar application, previously been submitted to this or any other funding body?
Yes

If Yes, provide details
This application represents an evolved and strengthened submission following feedback on my previous NIHR application. I have carefully considered reviewers' comments and incorporated their suggestions to improve the clarity, focus, and impact of the proposal. The proposal has also been submitted to Barts Charity, with interview invitations expected by March.

I have refined the methods for Work Package 2, which aims to understand variations in medication optimisation in CKD and CVD care by subgroups. This will now be a cohort study examining time to optimisation across population groups to identify inequalities in implementing national guidance. This approach is more robust than a snapshot of prescribing and enables broader analysis.

Rather than creating a new decision model for economic analysis, I will use the validated SHARP CKD CVD model, developed from long-term RCT outputs. This model is being expanded for my proposal to enable analysis of CKD stages and the effects of newer medications on broader outcomes, including heart failure. This refinement ensures high methodological standards, addresses concerns regarding data sources, and allows a focus on generating actionable insights instead of model development.

North East London (NEL) offers a socioeconomically diverse population to explore variations in prescribing patterns and health outcomes. NEL provides a representative and practical population for economic analyses, with results applicable to other settings. Integrated Care Boards (ICBs) require local intelligence rather than relying solely on national CPRD data. My work provides a blueprint for ICBs to use their population data, assess inequalities, and develop cost models for targeted interventions. NEL data is essential for modelling in Work Package 3 and delivering the insights required for commissioners, rather than CPRD. Leveraging the same data for Work Package 2 avoids duplication and lays the foundation for the decision modelling by providing the baseline analyses of medication optimisation inequities. This approach enhances understanding of health inequalities and informs interventions to reduce disparities and improve equity in healthcare delivery.

I have clarified the characteristics of the CKD population and subgroups included in the study. The population comprises individuals with increased kidney risk as per the KDIGO guidance (coloured orange or red on heat map), i.e. a GFR <60 mL/min/1.73m² or an ACR >3 mg/mmol. The study focuses on analysing prescribing pattern changes and addressing disparities in prescribing practices. These insights will help mitigate healthcare inequities and improve access to effective treatments for underserved groups.

For the training plan, I have streamlined courses to a focused set aligning directly with the research aims. I have removed the overseas research visit, opting instead to foster greater collaboration within the UK through professional network expansion and online platforms for discussions and knowledge sharing. This enhances feasibility and cost-effectiveness while promoting international partnerships.

Through these revisions, the proposal is now more focused, rigorous, and aligned with NIHR expectations, increasing its likelihood of delivering meaningful research outcomes.

Contextual Factors
Please use this section to detail any factors that may have impacted the development of your research career or outputs from your research career to date. NIHR wants to know about any circumstances so that they may take them into consideration during the assessment of your application. Contextual factors may include:

- Career breaks due to parental leave, or periods of illness.
- Reduced time spent undertaking research activities (including research outputs), due to disability or to caring responsibilities. This may include physical or mental health related impairments or conditions, which have impacted your opportunities to undertake research activities
- Reduced opportunities for career support e.g., mentorship, and limited opportunities to undertake prior research and training.
- Impact of the COVID-19 pandemic on your research career

Please also use this section to detail any other factors that may have impacted your research career not listed in the examples provided. The impact on your career to date will be specific to your particular circumstances but

could include such impacts as limited opportunities to obtain grant funding or fewer publications. In general terms, contextual factors should be significant, and relevant. NIHR acknowledges that you may be reluctant, or uncomfortable disclosing relevant information that is sensitive. However, you should bear in mind that we are unable to take into account factors that you do not disclose. Please be assured that the information provided by you is sensitive and will be treated confidentially and in line with the General Data and Protection Regulations (GDPR).

4. Plain English Summary of Research

Background

Chronic Kidney Disease is a serious health issue that can lead to kidney failure and heart problems if not properly treated. If the kidneys stop working, the person will need to have treatment to replace the kidney function, either by a kidney transplant or dialysis. As well as kidney failure, people with chronic kidney disease are at greater risk of heart attacks, heart failure, strokes and damage to the blood supply in their legs and feet. Some people from different backgrounds are affected more. Older people, men and people from Black or South Asian backgrounds have a greater risk of kidney failure and heart disease.

A range of medicines are used by General Practitioners to prevent people from having worsening kidney disease or developing heart disease. They improve blood pressure control, lower how much protein is in the urine, improve how well the kidney filters the blood, lower cholesterol, and lower sugar in the blood. Some of the medications are new and the impact of combinations of these medicines is unclear. More research is needed to find out whether some groups of people are undertreated and the impact that best treatment could have on reducing inequalities, costs and hospital use. This is important as it would help doctors and patients make better decisions about which groups are the most undertreated and which medications to use together to best reduce the risk of disease based on personal characteristics such as age, sex and ethnic background.

Aim of the research

This study aims to investigate how age, sex, ethnicity, and socioeconomic factors influence the effects of medications on the kidneys and heart in individuals with chronic kidney disease. It will evaluate impact of key medications used alone or in combination. Additionally, the study seeks to estimate the health and financial implications of using these medications optimally in a diverse urban population like North East London. The research will provide insights that can contribute to more personalized and effective treatment strategies for patients, taking into account the diverse characteristics of the population.

Design and Methods

I will review the latest evidence to find out what we know about how these medications work in people from different backgrounds and when used together.

I will review the medical records of people with chronic kidney disease in North East London to find out if there are differences in how the key medications are provided to people with different backgrounds including on age, sex, ethnicity, or socioeconomic factors.

I will then estimate what the likely kidney and heart benefits would be if we were to give the medications to everyone who should be on them and calculate the costs of doing so, which includes the savings from reduced hospital care.

Patient and Public Involvement

This summary was written with a group of people with lived experience of kidney and heart disease who believe this is an important research topic, and who I consulted with in the design of this research. I will continue to work with this group of expert patient advisors to use their knowledge and experience of accessing health care to help shape and guide the research and develop accessible summaries of the results.

Dissemination

I will share this information via published papers, in conference presentations and at patient groups guided by the expert patient advisors. This knowledge will help doctors tailor treatments and give patients the right information about what works for them, to support shared decision making, improving their quality of life and reducing the differences some receive in health outcomes.

5. Scientific Abstract

Background

Chronic Kidney Disease (CKD) is a significant health concern, often progressing to end-stage renal disease and cardiovascular disease (CVD), particularly in those from Black or South Asian backgrounds. In addition to blood pressure control, renin-angiotensin system inhibition (RASI), and lipid-lowering therapy (LLT), several new interventions and medications are available to reduce the risk of CKD progression and for CVD prevention., including sodium-glucose cotransporter-2 inhibitors (SGLT2s), glucagon-like peptide 1 agonists (GLP1) and finerenone. However, these treatments are underutilized, with variation for those from different backgrounds.

Aim

In patients with CKD, evaluate the health and financial impact of reducing inequalities in medication Optimisation for CKD and CVD prevention in a diverse UK urban population.

Objectives

1. **Systematic review on the impact of BP thresholds, RASI, LLT, SGLT2s, GLP1s and Finerenone on renal function and CVD outcomes in patients with CKD by equity factors including age, sex, and ethnicity.**
2. **Determine the variation in the time to medication Optimisation in the North East London (NEL) population with CKD by age, sex, ethnicity, deprivation and co-morbidities such as CVD, Diabetes and Severe Mental Illness**
1. **Estimate the expected additional health benefit of medication Optimisation on CKD progression and CVD prevention in the NEL CKD population and quantify the economic impacts with analysis of subgroups from objective 2 using the SHARP CKD CVD model.**

Methods

1. Work Package 1 will be an evidence synthesis of the differences by age, sex and ethnicity on the impact of key interventions for chronic kidney disease on renal function and CVD outcomes.
2. In Work Package 2, a cohort study will assess the time to medication Optimisation in the NEL CKD population and statistically analyse differences by subgroup to assess health inequalities.
3. Work Package 3 will involve decision modelling using SHARP CKD CVD to determine the health and financial impacts of medication Optimisation. This will begin by determining the level of prescribing if trial evidence and guidance was applied in full to the population. Then the difference in current prescribing to the optimal level will be assessed, with subgroup analysis. The decision model will estimate health and financial impacts of medication Optimisation over five years and a lifetime horizon, including the financial benefit to reducing variation in prescribing by population group on healthcare costs.

Outcomes and Potential Benefit

This research will offer a detailed analysis of inequalities in the implementation of national guidelines on medication Optimisation. It will identify the financial investments required to address these gaps and project real-world health benefits for CKD and CVD across different subgroups. The study will quantify the benefits of medications, assess health disparities, and uncover variations in guideline application. Insights gained will inform clinical decision-making, support commissioning decisions, and guide the development of targeted interventions and personalized treatments. The findings will be disseminated directly to healthcare leaders and patient groups, and via academic journals and conferences.

6. Detailed Research Plan

The Problem Being Addressed

Chronic Kidney Disease (CKD) is a long-term condition marked by gradual kidney function loss, potentially leading to end-stage renal disease (ESRD) requiring dialysis or transplantation.(7) CKD often coexists with cardiovascular disease (CVD), with 40-50% mortality in CKD patients due to CVD, compared to 26% in those with normal kidney function.(7–9) Common causes include diabetes and high blood pressure. The link between cardiovascular, metabolic, and renal diseases underscores the need for a holistic, tailored treatment approach, particularly with new drugs offering benefits for both renal and cardiovascular health. (10)

Medication Optimisation ensures patients receive clinically and cost-effective medications at the right time, with active involvement in decision-making.(11) Clinical trials have shown the importance of optimising blood pressure, renin-angiotensin system inhibition (RASI), lipid-lowering therapy (LLT), SGLT2 inhibitors, GLP-1 agonists, and finerenone in improving outcomes for both CKD and CVD. (12,13) These treatments, recommended in NICE guidelines, are variably implemented in primary care. (12) Around 30% of CKD patients in England do not reach their target blood pressure, with disparities in RASI and LLT use based on age, sex, and ethnicity, indicating healthcare access issues, bias and discrimination.(14,15) Interventions are needed to address this variation, and estimating the health and financial benefits will help justify investments. Expert patients reported a lack of communication about the benefits of medications, especially in combination, which may hinder their use.

Why This Research Is Important

The NHS England Long Term Plan aims to prevent over 150,000 heart attacks, strokes, and dementia by 2029.(16) The Department of Health Major Conditions Strategy emphasizes prevention, a whole-person approach, and empowering patients with the right information to make healthcare decisions.(17) However, many patients are not receiving effective preventative medications recommended by national guidance. (14,15)

This research was developed with input from individuals with lived experience of kidney or cardiovascular disease. Feedback from the Renal Patient Forum and public discussions highlighted a strong focus on prevention. Many expressed disappointment at missing key medications that are now common in CKD and CVD prevention. They emphasized the importance of early, timely access to medications. Patients in my practice also wanted more information to make informed decisions, asking about the benefits of specific medications and the best combinations. This research aims to address these concerns and improve patient experience and outcomes.

There are 3.25 million people with CKD in the UK, expected to rise to 3.9 million due to ageing, diabetes, hypertension, and inequalities. The number of people on dialysis is projected to increase from 30,000 to 143,000 in the next ten years, with NHS costs around £6.4 billion. Productivity losses for individuals with ESRD and their carers are £372 million, excluding loss of utility. (18)

In England, approximately two million adults have CKD, with variation in incidence and treatment across different groups. North East London (NEL), home to around 2.4 million people, is ethnically and socioeconomically diverse, with 42% Black or South Asian, and has four of the top six most deprived boroughs in London.(19) CKD and CVD outcomes are worse in Black, South Asian, and deprived populations. (14,20) NEL has 70,000 CKD patients, 24,000 of whom are from Black or Asian backgrounds. (14) Local data shows 26% of CKD patients in NEL are not on RASI, 50% on statins, and fewer than 30% on SGLT2s. This makes NEL an important setting for investigating health inequalities related to ethnicity and deprivation, particularly in CKD.

Worldwide, CKD affects 9.1% of the population (700 million people) and is the 12th leading cause of death, particularly impacting low- and middle-income countries. (21) A realist evaluation in a diverse, deprived area of London will provide insights into the global health benefits of expanding access to these drugs in resource-limited populations.

Review Of the Existing Evidence

To support this proposal, I conducted keyword searches on chronic kidney disease, cardiovascular disease, and inequalities, in PUBMED, MEDLINE and the Cochrane Library to find randomised controlled trials (RCTs) and systematic reviews evaluating the interventions.

Evidence for medications in CKD progression and CVD prevention

Blood pressure control and lipid-lowering therapy are key to CVD prevention. The Blood Pressure Lowering Treatment Trialists' Collaboration found a 5mmHg reduction in systolic BP lowers the risk of major cardiovascular events by 10%.(22) A 1mmol/L reduction in LDL-C with statins reduces ischaemic heart disease and stroke risks by 24%-25%, with larger reductions from intensive statin therapy.(23) In the SHARP trial, reducing LDL-C by 0.85mmol/L in CKD patients led to a 17% reduction in major atherosclerotic events, mostly due to fewer ischemic strokes and arterial revascularizations.(24)

RASI is central to CKD treatment in international guidelines, with strong evidence showing it slows GFR decline and reduces proteinuria.(12,25,26) NICE recommends blood pressure control, lipid-lowering therapy, and RASI use for CKD patients to prevent progression and CVD events.(12)

NICE recommends a BP target of 140/90mmHg for patients under 80 with hypertension, or 130/80mmHg if they have significant proteinuria (ACR ≥70mg/mmol).(12) Both ACCORD and SPRINT trials showed that intensive BP control improved CVD outcomes, with temporary GFR reductions due to haemodynamic changes, which were reversible, and lower incident albuminuria.(27,28) Kidney Disease Improving Global Outcomes (KDIGO) guidelines support intensive BP control targeting a systolic BP of <120mmHg, as the CVD benefits outweigh the risk of kidney damage.(26)

SGLT2 inhibitors, initially developed for Type 2 Diabetes, are now prominent in CKD and CVD. (29–32) A review of 13 trials with 90,413 patients showed a 37% reduction in CKD progression and a 23% decrease in cardiovascular death or heart failure hospitalization. (29) SGLT2s are now recommended for both high-risk CVD patients with Type 2 Diabetes and those with CKD, regardless of diabetes status.(33)

Obesity is a significant contributor to CKD.(34) GLP1s, originally for Type 2 Diabetes, improve weight management and CVD outcomes. The LEADER trial showed liraglutide reduced cardiovascular events, mortality, and microvascular outcomes, particularly renal effects. (35) GLP1s also slow eGFR decline and diabetic kidney disease. (36) Both SGLT2s and GLP1s are recommended for CKD patients with Type 2 diabetes by American Diabetes Association (ADA) and KDIGO guidelines.(25,37)

Mineralocorticoid receptor agonists are used in CVD but can cause hyperkalaemia and sexual side effects.

Finerenone has a lower risk of these issues. The FIDELITY trial showed a 14% relative risk reduction for cardiovascular outcomes and 23% for kidney outcomes, with NICE, ADA, and KDIGO recommending it. (25,38–40)

Cook et al analysed CKD prevalence in Type 2 Diabetes patients and treatment patterns over three years, finding low SGLT2 use and proteinuria measurement. (41)Further analysis on prescribing patterns across CKD populations and equity groups could improve outcomes.

Evidence of differences in CKD and CVD Risk

Certain populations face higher CKD and CVD risks. GFR declines by 10 ml/min per decade after age 40, amounting to a 30 ml/min drop by age 70. (37) Non-white populations experience greater renal decline.(42) Older, Black, and South Asian patients are more prone to hypertension, diabetes, and hypercholesterolaemia. (20) South Asians have higher risks of ischaemic heart disease, while Black people face higher stroke risks. (43)

Women are more likely to have CKD but experience slower progression than men, who have higher ESRD and CVD mortality risks.(44) Women, however, report worse quality of life, greater symptom burdens, and face barriers to kidney transplantation.(45) They also face increased CVD mortality risk at lower systolic BP than men.(46)

CKD is more prevalent in deprived populations, with faster progression, delayed presentation, higher CVD risk, and premature mortality.(47) However, limited studies and methodological gaps highlight the need for further research on deprivation's impact on CKD.

Medication usage and impact in different groups

Women are less likely to receive RASI and SGLT2s and show less heart failure risk reduction with SGLT2s (45,48,49) Black patients with hypertension are less likely to have controlled BP but no difference in treatment intensity.(3) Statin prescribing is lower in non-White patients with diabetes, requiring investigation in CKD (50). While U.S. studies explore ethnic differences in SGLT2 and GLP1 use, UK research is limited.(48,50–54) SGLT2s and GLP1s show significant ethnic variation in CVD and CKD effects in patients with Type 2 Diabetes, with limited benefit in Black populations.(53) An English Primary Care study linked general prescribing rates to deprivation for 65% of drugs.(55) Further research is needed to assess underrepresentation in trials or gender and ethnic variations in medication effects.(50,56)

Evidence for the use in combination

The key medications affect different parts of the prevention pathway and in theory, could be complementary.

Trials on SGLT2s have been as an add on to RASI. (25,32,35,36) GLP1s primarily benefit atherosclerotic events including stroke, myocardial infarction, and CVD mortality, but not heart failure. SGLT2s decreased risk of death and heart failure by 40%. Theoretically, combining GLP1 and SGLT2 could be synergistic.(57) CONFIDENCE is assessing the combined SGLT2 and finerenone impact on ACR. (58)

Cost Effectiveness Studies

Wu et al found that optimising blood pressure in inner NEL could yield 22,228 additional life years and save £41

million in hospital costs, while statin Optimisation could add 4,034 life years and save £12 million (59). SGLT2s and GLP1s are cost-effective for diabetes and related CKD and CVD complications.(60) However, NICE highlights a lack of economic analyses on these medications in CKD, CVD prevention, diverse populations, and combined use.(38)

Summary

CKD is a major health concern, with evidence showing that medication Optimisation reduces CKD and CVD outcomes. However, uptake remains poor, with disparities in treatment by ethnicity and sex. Granular evidence is needed to explore inequities across age, sex, ethnicity, and social groups. This proposal will analyse prescribing patterns in NEL, compare them to optimal levels based on national guidance, and quantify the gap between current practices and potential improved outcomes. It will estimate the health and financial benefits of medication Optimisation, including reduced cardiovascular events, hospitalizations, and progression to end-stage kidney disease. Understanding these impacts in diverse populations is vital for identifying optimal treatments, supporting policy, and guiding personalized care. With Renal Replacement Therapy funding transitioning to Integrated Care Boards (ICBs), this research will help commissioners evaluate medication Optimisation benefits, target inequalities, and maximize population health impact. It will also assist patients in choosing treatments tailored to their needs, addressing disparities and advancing personalized care.

Aim

In patients with CKD, evaluate the health and financial impact of reducing inequalities in medication Optimisation for CKD and CVD prevention in a diverse UK urban population.

Objectives

1. Systematic review on the impact of BP thresholds, RASI, LLT, SGLT2s, GLP1s and Finerenone on renal function and CVD outcomes in patients with CKD by equity factors including age, sex, and ethnicity. **(Work Package 1)**
2. Determine the variation in the time to medication Optimisation in the North East London (NEL) population with CKD by age, sex, ethnicity, deprivation and co-morbidities such as CVD, Diabetes and Severe Mental Illness (**Work Package 2)**
3. Estimate the expected additional health benefit of medication Optimisation on CKD progression and CVD prevention in the NEL CKD population and quantify the economic impacts with analysis of subgroups from objective 2 using the SHARP CKD CVD model. **(Work Package 3)**

6a. Project Plan

Work Package 1: Evidence synthesis

Trials have demonstrated the benefits of medication in slowing CKD progression and reducing CVD risk, but evidence on effects across different population groups requires compilation and evaluation. I will collate and synthesize trial evidence to assess the impact of medications on CKD and CVD in different groups to optimise treatment strategies for diverse populations. Additionally, I will identify gaps in the evidence to inform future research. The review will support modelling potential health impacts, reinforcing the case for tailored interventions in medication Optimisation. A scoping search of PROSPERO found no similar recent or prospectively registered reviews. The review will set the scene for the modelling the potential long term health benefits of increasing medication Optimisation in diverse populations.

Study Design

A systematic review examining the differences by age, sex, and ethnicity in the association of BP control, RASI, LLT, SGLT2s, GLP1s and Finerenone (see appendix) on CKD and CVD in adults with CKD.

PICOS

Population: adults≥18 with CKD globally

Intervention: CVD and CKD risk reduction (RASI, LLT, SGLT2s, GLP1s and Finerenone)

Comparison: With or without medications, between them and by combination, and based on age, sex, and ethnic groups.

Outcome: BP control, serum cholesterol, estimated glomerular filtration rate (GFR), albumin-creatinine ratio (ACR), Body Mass Index (BMI).

Studies: Primary: Randomised Controlled Trials, Systematic Reviews. Secondary: Cohort studies

Methods

Data Sources, Selection Criteria, & Search Strategy

I will be guided by Cochrane for robust methodology and PRISMA-P on the development of a protocol. To prevent duplication, I will register this systematic review with PROSPERO. Alongside an independent reviewer, I will conduct study selection, data extraction, and quality appraisal using a standardized data extraction form. Uncertainties or discrepancies will be resolved by a third independent reviewer. I will begin with a comprehensive search using predefined criteria in databases such as PubMed, Cochrane Library, and Embase, developing a search strategy using keywords.

Quality Assessment

I will assess the reporting of randomized controlled trials and cohort studies using the CONSORT and STROBE checklists respectively. Risk of Bias will be assessed using the Cochrane Collaboration's tools on observational and randomized trials.

Data Synthesis

Findings will be synthesized for exploration of differences in CVD and CKD impact and gaps in the literature. Key information under five headings (intervention description, causal assumptions, implementation, impact mechanism, and outcomes) will be extracted to describe the interventions. Forest plots and meta-analyses using STATA/R will compare mean differences and confidence intervals between intervention and control groups. Subgroup analyses will uncover variations based on demographic and socioeconomic factors to explore health inequality impacts. Sensitivity analysis will assess robustness by excluding studies with high bias or small sample sizes.

PPI Involvement

Patients wanted more information on the medication and their impacts on people like them for informed decision making. I will establish a core PPI group who will provide expertise and contextual information to support developing the protocol, analysing and interpreting outputs, and creating summaries.

Anticipated Outputs

- Comprehensive assessment of evidence base on age, sex, and ethnic differences in the association of BP threshold, RASI, LLT, SGLT2s, GLP1s and Finerenone on CKD progression and CVD prevention
- Assessment of evidence gaps and areas for further research
- The extent combinations have been evaluated for their effect on CKD and CVD
- Lay summaries to communicate evidence of differences in populations on the impact of medication optimisation

Work Package 2: Cohort characterisation

In the UK, older adults, males, and individuals of South Asian or Black African/Caribbean descent face a higher risk of adverse CKD and CVD outcomes. Timely, appropriate treatment informed by national guidance can significantly reduce these risks. Quantifying differences in the initiation and intensification of treatment in patients with CKD by equity groups will assess health inequalities in the implementation of national guidance. This comparison will support commissioners in the development of services to undertake targeted improvements. It will provide community group leaders with the information to work with their population on focused medication Optimisation.

Study Design

Observational cohort study comparing time to medication Optimisation in different population groups with CKD using NEL primary care data, to assess the variation in implementation of national guidance.

Study Population and Inclusion/Exclusion Criteria

Adults aged >/= 18 years old with diagnosed and coded CKD, registered with a NEL primary care practice at the time of data extraction, using an index date of 1st April 2025.

Variables/Covariates

Diagnosed CKD: defined as coded entry for increased kidney risk as per the KDIGO guidance (coloured orange or red on heat map), i.e. a GFR <60 mL/min/1.73m² or an ACR >3 mg/mmol.

Demographic Information: Data will be collected on age, sex, ethnicity, and deprivation. Deprivation scores will be obtained by using the Index of Multiple Deprivation (IMD). Self-reported ethnicity will be identified using SNOMED codes and grouped into the five main categories of the 2021 UK census (White, South Asian, Black African/Caribbean, Mixed, and Other).

Clinical Metrics: Blood pressure, GFR, ACR, Cholesterol, HbA1c, BMI

CVD Risk: Defined as QRISK2/3 level

Comorbidities: Hypertension, CVD, Diabetes, Severe Mental illness, and Learning Disability

Prescriptions: The key CKD and CVD medications prescribed in the previous 6 months.

Medication Optimisation

Optimisation is achieved when a patient is prescribed the medication recommended by guidelines and based on their co-morbidities and clinical metrics. At the start of the work package, the latest guidelines will be reviewed to identify the optimal medication regimes. They can be unoptimised, partially optimised or fully optimised. Partially optimised occurs when some appropriate medications are missing or not titrated to maximum dosage. Individuals

will experience periods of both optimised and non-optimised medication as their health profile changes. The time to Optimisation for each health state will be aggregated to account for this variation.

Data Source

I will use the Discovery Database, containing deidentified electronic health records for all 2.4 million patients in 266 NEL GP practices, including over 70,000 with CKD. NEL has a socioeconomically diverse population with 98% ethnicity recording in primary care. CVDPrevent analysis shows higher CKD testing and coding rates in NEL compared to other ICBs.(14) This enables detailed equity group analysis of medication Optimisation and health and financial impacts, which can be extrapolated and applied to other ICB populations to aid in their policy decisions. De-identified data based on SNOMED codes will be extracted using searches created by myself to be run in the Discovery Database. The data will be processed and stored in a data safe haven.

Index Dates

For the time from diagnosis to initiation of primary medication, baseline is defined as the date of coding CKD. For combination therapy, the baseline date is when the primary medication is initiated and clinical eligibility is reached for the additional medication. For example, the time to SGLT2 initiation is taken from the time at which the person has Type 2 Diabetes, has a GFR between 25-75 mL/min/1.73m² and is on RASI. Baseline covariates were identified based on the value closest to the date of CKD diagnosis or the date of medication eligibility, using data from the 12 months prior to or 3 months following these events. Prescribing data will include the type and dosage of current medications prescribed within the previous 6 months.

Data Analysis

Multilevel multivariable proportional hazard models will be used to identify differences in time to initiation and Optimisation of treatment. I will account for confounders and effect modifiers identified a priori via a directed acyclic graph, using multivariable Cox proportional hazards regression.

Additionally, the overall level of prescribing of each of the key medications at the index date will be reported and categorized as unoptimised, partially optimised or fully optimised. Multinomial logistic regression will be conducted to examine the differences in level of optimisation by the demographic factors and co-morbidities reported as odds ratios.

STATA/R will be used for statistical analysis. Sensitivity analysis will be undertaken using the date of the first low GFR and/or high ACR as the baseline instead of the date of diagnosis.

PPI Involvement

The data and analysis need to be representative and will be guided by the PPI group. They will be involved in defining the population group and reviewing the results. They will assist in preparing reports and their dissemination in NEL and nationally.

Anticipated Outputs

- Establish the current status and variation of medication optimisation
- Enhanced understanding of implementation of national guidance, prescribing patterns and health

outcomes by population group.
- Lay summaries on health inequalities in medication optimisation

Work Package 3: Modelling Study

Commissioners are soon taking over budgetary responsibility for renal replacement therapy from central NHS England. Projecting optimal medication use from trial data and national guidance to a real population would help estimate the impact on reducing need for replacement therapy, and narrowing inequalities, and the medication cost in comparison to the financial and health impacts. This will allow comparisons against other interventions and enable judgments on investment. This work package aims to offer insights to guide policy decisions on medication Optimisation and combination therapy through economic modelling. It will analyse the impact in 5 parts:

1. Determining the level of prescribing reached if the eligibility criteria from trial data and national guidance were applied in full to the NEL population.
2. Comparing the difference in the current level of prescribing to that reached if the eligibility criteria were applied.
3. Quantifying the level of health benefit achieved if optimal prescribing were reached at 5 years and over the lifetime.
4. Estimating the health and financial impact of improving the medication Optimisation for 10% of the population compared to the entire population
5. The optimal combination of medications from a health and cost perspective, if feasible comparing the additional health and financial impact of each intervention.

Commissioners have requested modelling the impact for 10% of the population as it aids development of business cases for improvement pilots.

This study will use the SHARP CKD-CVD decision-analytic model to evaluate the health and financial impacts of optimising medication for CVD prevention and chronic kidney disease CKD progression in NEL. Built using data from 9,300 participants in the Study of Heart and Renal Protection (SHARP), the model projects lifetime risks of cardiovascular events, CKD progression, survival (adjusted for quality of life), and healthcare costs. It is ideal for this analysis due to its ability to account for continuous exposures and the high comorbidity burden in CKD patients, where cardiovascular complications are a leading cause of mortality. Developed in collaboration with SHARP clinicians and epidemiologists, it integrates patient demographics, lifestyle factors, clinical characteristics, and evolving health histories. The model's performance has been validated against three external patient cohorts, ensuring robust risk prediction and applicability in guiding clinical and policy decisions. This model is being expanded to allow for greater analysis of CKD stages and the effects of newer medications on broader outcomes, including heart failure. I have access to the model code and will adapt it for use with help of the model author, who is a collaborator, and my health economist supervisor The analysis will compare current prescribing with optimal levels as defined by clinical guidelines and trial evidence, focusing on disparities across subgroups, particularly those affected by intersectional inequity and multimorbidity.

A 5 year and a lifetime time horizon will be adopted, with outcomes expressed as incremental cost-effectiveness ratios (ICERs), following NICE health economic analysis guidelines. Costs will be considered from the UK NHS perspective, while patient outcomes will be assessed through a lens of health equity and quality of life. Probabilistic sensitivity analysis will estimate uncertainty in key parameters using Monte Carlo simulation methods, generating cost-effectiveness acceptability curves to represent uncertainty across thresholds. A 3.5% discount rate will be applied to both costs and outcomes.

<u>Data Analysis</u>

Using NEL's adult CKD population (diagnosed and suspected), patients eligible for key medications (e.g., RASIs,

SGLT2 inhibitors, and statins) will be identified based on predefined clinical criteria from guidance and trial data. Individuals will be categorised into optimal, suboptimal, and untreated groups based on medication prescriptions in the previous six months. Projected prescribing levels will be compared to current practices to assess the proportion of optimally and sub-optimally treated patients, stratified by clinical and demographic factors.

Projected impacts on CKD progression (e.g., GFR and ACR) and CVD outcomes (e.g., BP, cholesterol, and events) will be analysed, along with mortality and quality of life. Both short- and long-term outcomes, including dialysis, and medication costs, will be reported.

PPI Involvement

Patients are keen to know what health benefits are of these medications and what the best combinations of medications would be for them. The Core PPI group will provide contextual information in shaping the modelling and analysis of benefits, interpreting the financial benefits from a patient's lens and helping to develop lay materials.

Anticipated outputs

- Quantification of optimal prescribing levels and expected health benefits over five years and a lifetime.
- Subgroup analysis highlighting variations in medication use and corresponding impacts on health and costs
- Evidence to guide investment and resource allocation for decision-makers and clinicians
- Create model of analysis to be undertaken by other Integrated Care Boards (ICBs) and policy makers to estimate the budgetary benefits of giving these drugs
- Accessible information for patients, including lay summaries and infographics, to support informed decision-making.

By quantifying potential reductions in cardiovascular events, kidney failure, hospitalizations, and mortality, the study will provide actionable insights to help commissioners evaluate cost-effectiveness and prioritize high-value interventions.

Dissemination, Outputs and Impact

This research will produce evidence on age, sex, and ethnic differences in the effects of key medications for CKD and CVD, project the benefits of optimal prescribing, and assess associated costs.

Outputs: WP1 will comprehensively assess subgroup differences in medication effects for CKD and CVD, highlighting evidence gaps and future research priorities. WP2 will analyse current patterns and variations in medication Optimisation by population group, evaluating the implementation of national guidance and differences in time to Optimisation. WP3 will quantify optimal prescribing levels, expected health benefits, and the financial impact of medication Optimisation, including the economic gains from reducing health inequalities. These outputs will support health service planning, inform policy and clinical decisions, and guide the development of targeted interventions and personalized treatments to reduce health disparities in CVD management for CKD patients.

Patient Dissemination: The dissemination strategy, co-developed with the PPI group, will ensure accessibility and relevance. Findings will be shared through community stakeholder meetings for direct impact and media engagement to broaden outreach. Plain-language summaries and visual aids, co-produced with the PPI group, will be tailored to engage specific communities effectively. Collaborations with Community Champions, organizations like the Community African Network and London Bangladeshi Health Partnership, faith-based groups, and the Barts Renal Patient Forum will further enhance dissemination efforts.

Commissioner Dissemination: Connections with London ICBs, the London Kidney Network, and Professor Nitsch's national links with the cardiorenal commissioner and UK Kidney Association will facilitate regional and national engagement. The research methods and findings will be shared with other ICBs, offering a blueprint to apply equity-group data and adapt the model for local populations to estimate the budgetary benefits of increasing prescribing.

Academic and Further Dissemination: I will disseminate my findings to the diverse range of stakeholder groups via peer-reviewed publications in high impact journals, presentations at national and international conferences, meetings with national policy stakeholders, and engagement with the media.

Short-term impact:

The research will identify subgroups with the greatest inequalities in medication Optimisation for CKD and CVD care. Evidence on these disparities will drive targeted improvements, supported by plain-language summaries and visual aids to ensure accessibility and engagement. Dissemination through community groups, faith-based organizations, and healthcare forums will maximize reach, while publications and conferences will advance clinical and research knowledge.

Medium-term impact:

Protocols, data analyses, and modelling tools developed during the study will be made freely available for further population health analyses and economic modelling across ICBs and other regions. Collaborations with London ICBs, the London Kidney Network, and regional renal networks will enable broader adoption of methodologies, supporting tailored interventions for diverse populations.

Long-term impact:

The findings aim to influence national kidney policy and inform the CORE20+5 program. The research model will provide a framework for assessing the health and financial benefits of addressing health inequalities in kidney care. Over time, it will enable the evaluation and refinement of services, driving significant reductions in disparities and improving outcomes for underserved and high-risk populations.

Risks

WP1 risk: The studies may not measure medication effects by equity groups sufficiently to assess differences in impact. A review of key studies, upcoming trials, and post hoc analyses indicates that evidence is available and ready for collation.

WP2 risk: Availability of data. The Discovery Database contains data on over 70,000 CKD patients, with nationally leading quality in ethnicity, CKD testing, and clinical coding. Access to this database is already in place and being used for service improvement.

WP3 risk: Dependence on WP1 and WP2 data. If medication impact data by equity group shows no variation, overall population trial data will be used. Baseline data from Discovery mitigates the risk of unavailable or insufficiently detailed input data. Decision modelling risk is addressed by using SHARP CKD CVD, a validated model with relevant transition probabilities, costs, and outcomes and a further update is due in time for the start of WP3.

Project management

The fellowship will be managed with support from my supervisors, who have significant experience in doctoral supervision. The GANTT chart outlines a reasonable and achievable timeline. For accountability, monthly meetings with updates, timeline reviews, progress reviews, and future work planning, along with regular email contact, are planned. Biannual progress meetings will be held with the patient working group. Wider institute and collaborator support will provide management, statistical assistance, and aid in dissemination. A data management plan will be approved by the Institute's Data Management Team. Financial management will be handled by the experienced finance department, proficient in managing grant funds, including NIHR and other funders' awards, and possessing extensive research-related financial management experience.

Ethics

The research uses secondary data and all GPs in North East London practices have consented to the use of their anonymised patient data for research and development for patient benefit. For rigor, ethical approval will be sought from the Queen Mary University Ethics of Research committee and NHS Health Research Authority. Good data governance will be ensured by implementing robust security measures to protect data from unauthorized access and following clear policies for data storage and retention.

Success criteria

Success will be through achievement of the specific aims and objectives supported by delivering at least three research papers. Active stakeholder involvement will be arranged, fostering meaningful connections, and ensuring relevance beyond academia including patients. Establishing a collaborator network will support professional relationships. Adherence to the GANTT chart timeline, and budget responsibility, following the financial breakdown, will be achieved with effective project management. The project will contribute to my skill development, as outlined in the training plan, offering hands-on experience and mentorship for professional growth. Overall, the project will be a foundational platform for my development and a career as an academic and clinical leader.

7. Patient & Public Involvement

Please describe how patients/service users, carers and the public have been involved in developing this proposal

A key priority at the outset of my research was to undertake a project in conjunction with members of the public rather than "to" them. I have identified core groups of interested and diverse patients who have been kindly supporting the development of this research. I have found their input to be invaluable in shaping acceptability and appropriateness of the research I have also sought guidance form the NIHR Research Design Service and INVOLVE which resulted in the following:

1. Outlined a clear rationale for PPI in my proposal
2. Creation of a strategy plan
3. Meet with a patient-participation group (Barts Renal Patient Forum)
4. Formation of a Core PPI group.

The plan prioritized:

1. early meetings to formulate the aims and objectives
2. clear communication in presentations and Lay summary
3. training development
4. resource planning
5. utilisation of findings to inform the application

2. **Outcomes from engagement**:

1) I have formed a core PPI group:

2) The Core PPI group have been involved

1. in shaping the direction of the research
2. have reviewed the application
3. helped shape the Lay Summary.

3) I have also presented to the Barts Renal Patient Forum. The key perspectives were:

1. Assessing the importance to patients of the outputs and risk early to enable understanding risk of adverse outcomes and benefits of treatment for important groups
2. Involvement in development of clear explanation and communication of what data is needed, how it is being collected, who is seeing it, how it is analysed and what will it be used for
3. Keeping in mind the applicability of how this data can then be used an aid to risk communication and decision making with patients
4. Consideration of Patient experiences of medication initiation and side effects.

4) Finally, I have discussed opportunistically with patients in my practice on their experiences of medicines Optimisation. They wanted more information to make decisions, including the benefits of the medications for individuals like themselves, and which combinations of medications had the greatest health impact

Please describe the ways in which patients/service users, carers and the public will be actively involved in the proposed research, including any training and support provided

Ongoing PPI input is crucial as it will support the understanding and development of risk assessment to better allow patients to make decisions about their health. The outputs will support clinicians to communicate treatment on a personalised basis.

I have formed a Core PPI advisory group of three diverse patient experts meeting bimonthly virtually or in person if possible.

They will:

- help further develop the study protocol
- suggest additional analyses
- contribute to report writing
- contribute to result dissemination – present results to their community organisations and groups and suggest ways of best doing so.

Undertaking Research

At each PPI meeting, the core group will provide expertise to develop the study protocol and as the project takes shape, ensuring oversight and steering the direction of the analysis. I will present the research design of each work package and current findings for members to advise on the direction of the work, key future priorities and additional ways they might be involved.

Report and Dissemination

Plans to report and disseminate the findings will be created via the PPI meetings and emails. The group will be involved in jointly creating lay summaries and infographics, which will be shared via the Patient Forum, the London Kidney Network, patient newsletters and social media platforms. I have links with the Community Champions, Community groups, local mosques, temples and churches, and the Barts Renal Patient Forum. I will ask stakeholders and interested parties to share findings via their networks or suggest how best to do this.

PPI Training and Development

Participants will be paid for meeting attendance and training as calculated by the NIHR Centre of Engagement and Dissemination and using INVOLVE guidance. I will make use of the existing PPI expertise in QMUL, in particular Dr Megan Clinch who leads the Public Advisory Panel, and the excellent resources and courses provided by the NIHR Cambridge Biomedical Research Centre. Members will be informed of their role and responsibilities including the time commitment and sent summaries of meetings. I will be available for questions and discussion outside of the meetings to provide updates or clarification.

If it is considered not appropriate and meaningful to actively involve patients/service users, carers and the public in your proposed research, please justify why
n/a

8. Training & Development and Research Support

Proposed training and development programme

This fellowship will provide the foundation to being an independent health services researcher and build upon the skills I have developed in my clinical roles. I want to build a portfolio of research and an in-depth understanding of how to apply medicine interventions to improve prevention and reduce risk of cardio-renal-metabolic disease in underserved populations. The **comprehensive training package** below has been designed to meet my learning needs, and give me research, management and leadership skills. I have planned a blended learning approach with intensive short courses, spread evenly throughout the duration of the fellowship with on-the job training, bespoke tutoring and international visits. The training is designed to introduce me to a breadth of disciplines, institutions, and people, which will help develop new ideas, methods, and collaborations, and to enhance my development as a clinical academic and future leader.

The training package will be overseen by my **highly qualified and experienced supervisors and advisory panels**. Queen Mary University has a strong history of nurturing future leaders and has the academic environment necessary for successful researchers to build high quality skills and expertise. This QMUL Researcher Development Unit provides a comprehensive spectrum of courses and a range of support for PhD students across the four UKRI domains: Knowledge and intellectual abilities; Personal Effectiveness; Research Governance and Organisation and Engagement; influence and impact. These include presentation, critical appraisal, grant writing skills, coaching, project management, public engagement and communication skills.

Pharmacoepidemiology

Causal Inference in Epidemiology: Recent Methodological Developments course is essential for gaining a foundational understanding of distinguishing causal inference to be able to employ the advanced analytical methods needed on the data and determine the real-world cause and effect relationship of the medications on CKD and CVD.

Pharmacoepidemiology & Pharmacovigilance and Real-World Evidence in Pharmacoepidemiology courses will equip me with the knowledge and skills to generate real world evidence on effectiveness of the medicines on CKD and CVD and overcome common sources of error, using electronic health record data.

Economic Evaluation and Decision Modelling

The **Statistical Methods in Economic Evaluation course** will provide an understanding of the fundamental statistical concepts used in economic evaluation and how to apply and interpret the results of advanced techniques for the analysis of data collected via observational studies.

Decision Modelling for Health Economic Evaluation Online Courses is designed to support building, interpreting and appraising decision models. Here, I will develop skills and knowledge in decision analysis for projecting the outcomes and effects of medications.

Evidence Synthesis

I have experience in using research to answer questions and apply evidence to clinical and policy questions. To support the structured collation, assessment and summarization of key evidence to apply to further develop research, I will undertake the **Systematic Reviews and Meta-analyses of Health Research** course.

Healthcare Data Science

The overseas research visit will provide me with exposure, knowledge transfer and data science skills from an internationally acclaimed and leading health data institute. I will use the support of my supervisors and the department at QMUL who deliver the Health Data in Practice MRes. I will have access to and can audit the modules on **Health data in Practice, Data Mining** and **Effective and efficient evaluation**.

Leadership

While I have leadership experience from my other roles, I would benefit training with a research lens. Good leadership and management are crucial to deliver high-quality research that is efficient, focused and delivers value for money. I will utilise the **NIHR Leaders Support and Development Programme**, which includes a range of

modules, workshops, courses and mentoring. Following an in-depth analysis, it was developed specifically for the NIHR to understand the steps to successful leadership and aims to support network building and the application of learning to real-time leadership.

Network Building and Research and Evidence Updates

Expanding my network will provide the opportunity to expand knowledge on the latest advances and innovations in research in cardiovascular disease and kidney disease. The interventions are relatively new, and this is a rapidly developing area with new evidence emerging, so it is key to be able to keep up to date and network with key researchers. I have a diverse stakeholder group in health care commissioning through my contacts in ICBs in London, the London Kidney Network, the Cardiac Operational Delivery Network and via my supervisors who have links to the cardiorenal commissioner. I also have international collaborators who are involved in health policy and clinical trials who can provide input and updates on medications and help in realizing the impact.

Inequalities

To further my understanding of health inequalities, I will complete the **Online Inequality in health and health care** course from Oxford University. The course covers theoretical concepts and an overview of household survey data and cohort studies which combine socio-economic and demographic variables with health measures. It will introduce me to diverse socio-economic issues in health and the role of research and expand the horizons of this and future research.

Academic Writing and Communication

The UK EQUATOR Centre's Publication School course will provide skills in academic writing and publication, in particular writing with simplicity and effectiveness and dealing with peer review.

Patient and Public Involvement

I am keen to engage patients in the guidance of the project and in developing the dissemination. I will make use of the expertise in QMUL, in particular Dr Megan Clinch who leads the Public Advisory Panel, and the excellent courses provided by the Cambridge Biomedical Research Centre. which will help me to involve the public in research, form a PPI group, involve them in study design and analysis, write lay summaries and in public communication.

RESEARCH SUPPORT / SUPERVISOR 1

Name	Professor Rohini Mathur

Institution	Queen Mary University

Position	Professor and Chair of Health Data Science

Description of support provided
Professor Rohini Mathur is Professor and Chair of Health Data Science in the Centre for Primary Care at QMUL. She brings significant expertise in the use of electronic health and longitudinal biomedical and cohort data. She has developed ontologies, algorithms, and best practice guidelines for the optimal use of EHRs and linked data, lead research into ethnic inequalities in trajectories of chronic disease care and outcomes and contributed to the development of healthcare policy and clinical guidelines and furthered our scientific knowledge about the aetiology of chronic conditions in diverse populations. Professor Mathur has particular interest in pharmacoepidemiology research to inform best clinical practice, guidelines, and policy. She will provide guidance in the creation, management, and analysis of complex electronic health databases and linked data sources and advise on statistical analysis including the development of scientific and ethical protocols, development of code lists, and methods of longitudinal analysis. I will also learn from her

expertise in health inequalities research and data analysis.

RESEARCH SUPPORT / SUPERVISOR 2

Name	Professor Jianhua Wu

Institution	Queen Mary University

Position	Professor of Biostatistics and Health Data Science

Description of support provided
Professor Jianhua Wu is Professor of Biostatistics and Health Data Science, and Centre Lead for the Centre for Primary Care at QMUL. His main research interests are statistical application in cardiovascular disease; data mining, machine learning analysis and methodological development for electronic health record data; data linkage for research-ready database, and new design and analysis for clinical trials. He has expertise in dealing with large cohorts and longitudinal electronic health records with advanced statistical and machine learning techniques. Professor Wu has substantial experience in interrupted time series analysis to assess the impact of intervention in clinical settings. He will provide guidance in the curation of the data in the electronic health database and provide expertise in the data collation and statistical analysis including the development of search protocols, codesets, and handling missing data.

RESEARCH SUPPORT / SUPERVISOR 3

Name	Professor Dorothea Nitsch

Institution	London School of Hygiene and Tropical Medicine

Position	Professor Of Clinical Epidemiology & Consultant Nephrologist

Description of support provided
Professor Dorothea Nitsch is Professor of Clinical Epidemiology at London School of Hygiene and Tropical Medicine (LSHTM) and Consultant Nephrologist at Royal Free Hospital. She is an expert in healthcare data analysis. Previously she was the analytic lead of the National CKD Audit which investigates the identification and management of CKD in primary care in the UK, has been collaborating with the UK Renal Registry since 2003 on various analyses related to the outcomes of patients on dialysis, and currently is the UK Kidney Association Director of Informatics Research. She is involved in research on risk factors and outcomes of chronic kidney disease and her interests include CKD risk factors and outcomes, life course epidemiology, cardio-renal-metabolic disease, and inequalities. Professor Nitsch will bring expertise in utilising big data, pharmaco-epidemiology, chronic kidney disease and healthcare analytics.

RESEARCH SUPPORT / SUPERVISOR 4

Name	Dr Runguo Wu

Institution	QMUL

Position	Senior Health Economist

Description of support provided
Dr Runguo Wu is a health economist focusing on the development of long-term cardiovascular disease models based on large clinical trials and population cohort data. His work involves the economic evaluation of healthcare interventions using these models. His research interests also include health inequity using data from routine health records, cohort studies, and clinical trials.

Collaborations
I have made use of my existing network and have sought new collaborations to support my development and to help guide the research. These cover three core areas: 1. Health Policy including Economic Modelling 2. Kidney disease and Cardiovascular disease 3. Health care data Professor Borislava Mihaylova is a health economist leading research on impact of diseases and treatments on morbidity, mortality, health-related quality of life and healthcare/other resources and costs; cost-effectiveness of prevention and treatment strategies. She develops applied and methodological research across high burden disease areas and public health problems to inform national and international health policy. She led the development of the SHARP CKD CVD model and is a co-investigator on the update of the model targeting SGLT2i and other forthcoming treatments. Professor Richard Grieve is a leading UK health economist and has significant experience of working in policy and for health care decision making. His research interests of decision modelling and evidence synthesis for health technology assessment align with the aims and methods of this project.. Dr Kieran McCafferty is a Consultant Nephrologist at Barts Health NHS Trust, Senior Lecturer and Queen Mary University London and Co-Chair of the Prevention Workstream of the London Kidney Network. He is divisional director of research for specialist medicine at Barts Health and the deputy clinical director of Research and Development. He will provide expertise into the synergy between kidney disease and cardiovascular prevention. Professor Riyaz Patel is Professor of Preventative Cardiology and Consultant Cardiologist at Barts Health and UCLH NHS Trusts. He will advise on cardiovascular risk. prevention, epidemiology and health informatics and provide his expertise on policy research. Professor Catherine Clase is professor of medicine in nephrology at McMaster University in Canada, and an associate member of the department of health research methods, evidence and impact. She will contribute to my understanding of population epidemiology of CKD and medication optimisation in CKD. Professor Juan Jesus Carrero is Professor in cardio-renal Epidemiology at the Department of Medical Epidemiology and Biostatistics, Karolinska Institute, Sweden. He will contribute to the use of health care data and the prevention and management of persons with CKD, and how to integrate kidney function assessment into the medical decision process. Dr Douglas Lee is a Professor of Medicine at the University of Toronto, Canada, and senior scientist and leader of the Cardiovascular Program at ICES. He will share his expertise in outcomes and health services research, health policy, and large database research. Within QMUL, I will work alongside healthcare data experts in the Clinical Effectiveness Group, who have over 40 years of experience of turning primary care data into clinical improvements. Their knowledge will help me in electronic health record data extraction, validation and analysis.

I have support from NHS organisations, including the London Kidney Network and the London Cardiac Operational Delivery Board, which collaborate across London with Integrated Care Systems, Public Health, Data Analysts, Commissioners, and patients. Their networks provide access to experts to guide the research, opportunities for wide dissemination, and experience in shaping policy and services.

NHS North East London ICB plans and commissions services to meet diverse population needs, ensuring system-wide effectiveness. My collaboration with them will focus on applying economic evaluations in health policy, enhancing patient engagement, and improving communication. They will benefit from the research outputs and my strengthened data and clinical expertise.

These collaborations will provide expertise on the clinical factors and the interplay between renal and cardiovascular disease, support my data and research skills, provide support for advancing my expertise in health policy, and complement my supervisors and their knowledge and skills.

Host Organisation Support Statement
The following statement of support must be completed by the head of your lead academic department. In order to contribute to this application, this individual must be invited to do so via the 'Participants and Signatories' section of the form.

Statement of support

As Deputy Director of Queen Mary University of London's (QMUL) Wolfson Institute of Population Health (WIPH), situated in the Faculty of Medicine and Dentistry, I am delighted to offer my strongest support for this application.

Dr Chris Carvalho is an experienced GP and a Research Fellow at our Centre for Primary Care's Clinical Effectiveness Group undertaking population health data analysis and quality improvement. He combines his clinical experience and analytic skills to research and publish on case-finding of familial hypercholesterolaemia, COVID prevalence in ethnic groups, and inequities in hypertension management. He has synthesized evidence for a clinical guideline on Sodium-Glucose Co-transporter-2 Inhibition in adults with kidney disease and the use of primary care databases for research. This fellowship is the natural next step in Dr Carvalho's career, following an NIHR School for Primary Care Research Career Progression Fellowship, and drawing on his policy role in NHS North East London Integrated Care Board.

This is a refined and strengthened proposal, building on an earlier submission that reached interview stage. Dr Carvalho has worked closely with his supervisors, engaged with diverse stakeholders, and thoughtfully integrated their feedback. The result is a robust and impactful research application, designed to deliver locally relevant insights for the population he sees clinically in North East London. In addition, it provides the blueprint for modelling the health and financial impacts of medication optimisation for diverse populations in other areas nationally.

This fellowship will be based in WIPH's Centre of Primary Care, and his training overseen by Professors Rohini Mathur and Jianhua Wu. The Centre for Primary Care is a member of the NIHR School for Primary Care Research (SPCR). The SPCR focuses on facilitating high quality primary care research through strategic leadership and ringfenced funding, and building capacity in primary care research through an established training programme. Chris will have access to SPCR training resources, networks and opportunities across the nine HEI members of the SPCR and will have strong links to the NIHR Schools for Social Care Research and Public Health Research. Our research encompasses a wide range of qualitative and quantitative approaches with research themes including wider determinants of health, quality improvement and health inequalities. The Centre for Primary Care runs the Health Data Science Seminar Series and the Curiosity Workshops.

The WIPH currently hosts 130 postgraduate research students, with substantial support for further recruitment. We run two Wellcome Doctoral Training Programmes (Health Data in Practice, and for Primary Care Clinicians), and benefit from additional programmes led from other Faculty institutes: the Programme for Health Advances in Underrepresented Populations and Diseases, and the Programme for Global Health Research in Africa. The WIPH also recruits doctoral students via the SPCR PhD Studentship scheme, the MRC/NIHR Trials Methodology Research Partnership, the London Interdisciplinary Social Sciences (LISS) doctoral training partnership, and City

of London Cancer Centre PhD studentships, as well as other national and international schemes.

Academic support for WIPH postgraduate research students is led by their PhD supervisors, and their journey from enrolment to graduation is overseen by our co-Directors for Graduate Studies and one professional services staff member- this includes signposting to QMUL central resources around wellbeing, travel awards, training and educational events. Given the volume and breadth of early career researchers within the Institute, they have formed their own active Early Career Researcher (ECR) group, which the applicant will also join. The ECR group receives Institute funding for bespoke training and/or expenses for invited external speakers.

In addition to development opportunities within the Centre of Primary Care and WIPH, Dr Chris Carvalho would also have the opportunity to acquire transferable skills throughout the Fellowship. Within QMUL, the Centre for Academic and Professional Development provides a comprehensive programme of short courses and bespoke support from which students from any Faculty may select an appropriate programme. These include generic writing and presentation skills, personal organisation, specific skills (e.g., using reference management software), career planning, and dealing with the media. Where appropriate formal transferable skill training is not available within QMUL, Dr Chris Carvalho will be supported to access appropriate training informally or elsewhere. QMUL's Centre for Public Engagement also offers a number of funding avenues to develop public engagement and PPI work. For example, a monthly small grants scheme (up to £600) supports projects that are engaging patients, service-users, or members of the public with health-related or engagement-led research.

In summary, QMUL offers an unparalleled environment within which to undertake this training. QMUL's research outputs, impact performance and research environment were all highly competitive as assessed in the 2021 Research Excellence Framework. We were ranked 7th in the UK for quality. Within the Faculty of Medicine and Dentistry, in Clinical Medicine 87% of outputs were ranked 3*-4*, 97% of the impact statements were ranked 3*-4*, and 100% of the environment was ranked 3* - 4*. In Public Health, Health Services and Primary Care 89% of outputs ranked 3*-4*, and 100% 3*-4* in both environment and impact case studies.

QMUL is one of the most diverse, research-led universities in the world. Equality diversity and inclusion (EDI) principles are fully and actively embedded into all day-to-day and strategic decision-making within the Faculty of Medicine and Dentistry, and reflected in our research aims ('Every life is precious'). The overarching mission is to create a truly inclusive environment, building on our cherished cultural diversity, where students and staff flourish, reach their full potential and are proud to be part of the University. EDI is embedded into all meeting structures, forums and policies and procedures within each of the six FMD institutes, including WIPH. Faculty and Institutelevel leadership committees contain EDI representation, and institute EDI reps ensure that faculty-level initiatives result in meaningful outcomes as well as developing and leading local change.

In conclusion, the development of our early research staff is pivotal to the development of the research and training base of the University, the region, and the national research agenda. The University would be delighted to host this award.

9. Detailed Budget

The finance section should provide a breakdown of costs associated with undertaking the research as described in the proposal.

1

Staff type	Lead Applicant

Name	Christopher Carvalho

Employing organisation	Queen Mary University of London

Grade	C11SL

Starting point on scale	008

Increment Date	01/08/2025

Justification of Costs
The salary costs will be for me as Principal Investigator, working part time at 0.8WTE. I have checked my salary scale with the Finance committee and have matched my experience as GP of 15 years.

Details of Posts and Salaries Costs

	Total (£)
Salary (£)	473554
Geographical Weighting (£)	5513
Other allowances (£)	2396
Superannuation (£)	86529
Nat. Insurance (£)	69048
Current Annual Costs (£)	**637040**

Staff Type: (Lead Applicant)-Dr Chris Carvalho
Type of cost: HEI
% Full-time on this research: 80.0
Total months on this research: 45.0
Justification of Cost: These salary costs will be for me as Principal Investigator, working part time at 0.8WTE. I have checked my salary scale with the Finance committee and have matched my experience as GP of 15 years.

	Year 1	**Year 2**	**Year 3**	**Year 4**	**Total Cost (£)**
Basic Salary	126281	126281	126281	94711	473554
National Insurance	18413	18413	18413	13810	69049

Superannuation	18371	21926	26419	19815	86531
Geographical weighting / Other allowances (if applicable)	1470	1470	1470	1103	5513

	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
Total HEI	131628	134472	138066	103551	507717
Total NHS	0	0	0	0	0
Total Commercial	0	0	0	0	0
Total Other	0	0	0	0	0

Travel, Subsistence and Conference Fees

Type of cost: HEI
Description: Subsistence
Conference: False
Item Description: The UK EQUATOR Centre's Publication School course - Accommodation and subsistence
Justification of Cost: The UK EQUATOR Centre's Publication School course will provide skills in academic writing and publication, in particular writing with simplicity and effectiveness and dealing with peer review.

	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
	185	185	185	138	693

Type of cost: HEI
Description: Conference Fees
Conference: True
Item Description: Conferences
Justification of Cost: I plan to present at a selection of conferences including SAPC, RCGP, UK Kidney Week, American Heart Association and European Renal Association. I will determine which are best to present at based on timing of the conferences and relevance to the research at the time. These are important opportunities for dissemination in the UK academic primary care field and maximising the impact of my work, networking with other researchers in my field and engaging with the work of others closely aligned to my research interests.

	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
	800	800	803	599	3002

	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
Travel, Subsistence and Conference Fees Total	985	985	988	737	3695

Equipment
No costs have been added

Summary

	Travel (excl Conferences)	Travel (Conferences)	Equipment
Total HEI	554	3002	0
Total NHS	0	0	0
Total Commercial	0	0	0
Total Other	0	0	0

Consumables
No costs have been added

Patient and Public Involvement

Description: Core Advisory PPI Group
Type of cost: HEI
Please describe fully: Core Advisory PPI Group
Justification of Cost: As outlined in the application, patient and public involvement is pivotal to the eventual success of a research project and I have therefore factored this into the application with the costs of reimbursement costs for the attendees

	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
	960	960	963	718	3601

	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
Patient and Public Involvement Total	960	960	963	718	3601

Other Direct Costs

Description: Data Management and Storage Costs
Type of cost: HEI
Please describe fully: Data Management and Storage Costs
Justification of Cost: To ensure Good data governance I will need a data safe haven and data management costs. This includes implementing robust security measures to protect data from unauthorized access and following clear policies for data storage and retention.

	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
	185	185	186	138	694

Description: Discovery Data Access
Type of cost: HEI
Please describe fully: Discovery Data Access
Justification of Cost: For access to Discovery database, including running and processing of searches, data extraction and data management.

	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
	6531	6531	6531	4885	24478

Description: Publication Fees
Type of cost: HEI
Please describe fully: Publication Fees
Justification of Cost: To support the wider dissemination costs.

	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
	1067	1067	1070	798	4002

	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
Other Direct Costs Total	7783	7783	7787	5821	29174

Training And Development

Type of cost: HEI
Description: Applicant PhD Tuition Fees
Item Description: PhD Fees
Justification of Cost: The PhD application fee and course fees are necessary costs of undertaking a doctoral fellowship

	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
	5015	5015	5119	3818	18967

Type of cost: HEI
Description: Short Courses
Item Description: Courses
Justification of Cost: Systematic Reviews and Meta-analyses of Health Research Justification of Cost: I have experience in using research to answer questions and apply evidence to clinical and policy questions. To support the structured collation, assessment and summarization of key evidence to apply tofurther develop research, I will undertake the Systematic Reviews and Meta-analyses of Health Research course. Online Inequality in health and health care Justification of Cost: To further my understanding of health inequalities, I will complete the Online Inequality in health and health care course from Oxford University. The course covers theoretical concepts and an overview of household survey data and cohort studies which combine socio-economic and demographic variables with health measures. It will introduce me to diverse socio-economic issues in health and the role of research and expand the horizons of this and future research. Causal Inference in Epidemiology: Recent Methodological Developments Justification of Cost: Causal Inference in Epidemiology: Recent Methodological Developments course is essential for gaining a foundational understanding of distinguishing causal inference to be able to employ the advanced analytical methods needed on the data and determine the real-world cause and effect relationship of the medications on CKD and CVD. Pharmacoepidemiology Justification of Cost: Pharmacoepidemiology & Pharmacovigilance and Real-World Evidence in Pharmacoepidemiology courses will equip me with the knowledge and skills to generate real world evidence on effectiveness of the medicines on CKD and CVD and overcome common sources of error, using electronic health record data. Decision Modelling for Health Economic Evaluation Online Courses Justification of Cost: Decision Modelling for Health Economic Evaluation Online Courses is designed to support building, interpreting and appraising decision models. Here, I will develop skills and knowledge in decision analysis for projecting the outcomes and effects of medications. Statistical Methods in Economic Evaluation course Justification of Cost: The Statistical Methods in Economic Evaluation course will provide an understanding of the fundamental statistical concepts used in economic evaluation and how to apply and interpret the results of advanced techniques for the analysis of data collected via observational studies. UK EQUATOR Centre's Publication School Justification of Cost: The UK EQUATOR Centre's Publication School course will provide skills in academic writing and publication, in particular writing with simplicity and effectiveness and dealing with peer review. NIHR Future-Focused Leadership programme Justification of Cost: While I have leadership experience from my other roles, I would benefit training with a research lens. Good leadership and management are crucial to deliver high-quality research that is efficient, focused and delivers value for money. I will utilise the NIHR Leaders Support and Development Programme, which includes a range of modules, workshops, courses and mentoring. Following an in-depth analysis, it was developed specifically for the NIHR to understand the steps to successful leadership and aims to support network building and the application of learning to real-time leadership.

	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
	3017	3017	3026	2257	11317

	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
Training And Development Total	8032	8032	8145	6075	30284

Total Direct Costs

	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
Total Direct Costs	147603	150447	154162	115565	567777

Indirect costs

HEI Indirect costs
No costs have been added

Commercial Indirect Costs
No costs have been added

Other Partner Organisation Indirect costs
No costs have been added

	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
Total Indirect Costs	0	0	0	0	0

Research Costs Requested From Funder

	Direct costs	Indirect costs/Overheads	Amount requested
Total Higher Education Institution Costs (80%)	534491	0	534491
Total Higher Education Institution Costs (100%)s	33286	0	33286
Total Commercial Costs	0	0	0
Total Other Parther Organisation Costs	0	0	0
Total NHS Research Costs	0	0	0
Total Research Costs Requested From Funder	567777	0	567777

Summary

	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
Salaries	**131628**	**134472**	**138066**	**103551**	**507717**
Travel-subsistence-and-conference-fees	**948**	**948**	**951**	**709**	**3556**
Equipment	**0**	**0**	**0**	**0**	**0**
Consumables	**0**	**0**	**0**	**0**	**0**
Patient-and-Public-Involvement	**768**	**768**	**770**	**574**	**2880**
Other-Direct-Costs	**6227**	**6227**	**6230**	**4656**	**23340**
Training And Development	**8032**	**8032**	**8145**	**6075**	**30284**
HEI-Indirect-costs	**0**	**0**	**0**	**0**	**0**
Commercial-Indirect-Costs	**0**	**0**	**0**	**0**	**0**
Other-Partner-Organisation-Indirect-costs	**0**	**0**	**0**	**0**	**0**
Total	**147603**	**150447**	**154162**	**115565**	**567777**

NHS Support Costs
No costs have been added

Have you discussed and agreed these costs with the Lead Network?	N/A

NHS Treatment Costs
No costs have been added

Have you discussed and agreed the NHS costs with the Lead Trust?	N/A

Is the patient/service user care being provided different from the usual standard treatment for the condition?
N/A

Usual Treatment Costs
No costs have been added

Excess Treatment costs/(savings)

	Cost per patient(£)	Year 1	Year 2	Year 3	Year 4	Total Cost (£)
Total NHS Excess Treatment costs/(savings)	0	0	0	0	0	0

	Total (£)
NHS Support Costs requested from networks	0
NHS Excess Treatment costs/savings	0
TOTAL NHS SUPPORT AND TREATMENT COSTS	0

Total Funding Required (including non-research costs)

	Total (£)
Total Research Costs Requested from Funder	567777
Total NHS (Support and Excess Treatment) Costs	0
Total Funding Required	567777

10. Uploads

Please upload the following documents, if necessary.

When uploading a document to your application, please ensure that it is in an appropriate file format. It is not possible to upload JPEG images, and with the exception of the SoECAT Form, which should be uploaded as a .csv file or Excel spreadsheet, all files should be uploaded in Adobe PDF or Word DOC/DOCX format. Please refer to the Guidance Notes for further information on the attachments required.

(References - NIHR CKD CVD 25 references.docx) is included as an appendix within this file.

(Figures / Tables - Appendix.docx) is included as an appendix within this file.

(Research Timetable - NIHR PhD 25 Timetable - Gantt Chart -final.pdf) is included as an appendix within this file.

(SoECAT Template - SoECAT Requirement Justification.pdf) is included as an appendix within this file.

11. Participants and Signatory

Participants
New participants must confirm their participation on your application before you will be able to select the 'SUBMIT' option. Please see the Submission Process Flow Diagram and the Applicant Guidance Notes for further information.

You are required to supply the names and email addresses (if not already registered with this online application system) of each required participant. Everyone **newly** named in this section will be acting as a 'participant' to your application and will need to agree to be part of this application.

By confirming participation, participants are acknowledging their involvement and input into this application, and agreeing to be involved in the proposed fellowship. You must ensure all **new** participants are happy for your application to be submitted before submitting it on the online system.

It is permissible for a single individual to serve as a signatory and a participant, but they will need to be added separately under each heading.

Details of the individual who will be required to approve your application after submission should be entered in the 'Participants and Signatory' section. Please note that it is acceptable for a Head of Department to also act as a research support role. If this is the case, they will need to be included in this section for both roles.

Participants must confirm their participation on your application before you will be able to press the submit button. Please see the 'Application Submission Process Flow Diagram' in the guidance notes for further information.

Doctoral Primary Supervisor

Doctoral Primary Supervisor Participant

Title	Professor
First Name	Rohini
Last Name	Mathur
Position	Professor and Chair of Health Data Science

Supervisor

Supervisor Participant

Title	Professor
First Name	Dorothea
Last Name	Nitsch
Position	Professor of Clinical Epidemiology and Honorary Consultant Nephrologist

Supervisor Participant

Title	Professor
First Name	Jianhua

Last Name	Wu
Position	Professor of Biostatistics and Health Data Science

Supervisor Participant

Title	Dr
First Name	Runguo
Last Name	Wu
Position	Senior Health Economist

Administrative Authority or Finance Officer

Administrative Authority or FO Participant

Title	Mrs
First Name	Coleen
Last Name	Colechin
Position	Operations Manager

Signatory
Participants must complete their actions on your application before submission; the signatory must also approve your application after you select the 'submit' option but BEFORE the application submission deadline.

You are required to supply the name and email address (if not already registered on the ARAMIS application system) of the individuals who will be 'signing off' your application. Once their contact details have been entered below, these individuals will be invited to log in and confirm their participation in your application.

The signatory must ALSO approve the application AFTER you have selected the 'SUBMIT' option but BEFORE the application submission deadline. Please see the Submission Process Flow Diagram and Applicant Guidance Notes for further information.

The signatory approval will result in the application being fully submitted to the NIHR. All parties (applicant, participants and signatory) will be notified of this via an automated system-generated email.
NIHR will not accept any applications unless fully approved by your signatory before the 1:00 pm deadline on Thursday 16th January 2025.

Host Organisation - Heads of Department

Head of Department or Senior Manager

Title	Professor
First Name	claudia
Last Name	cooper

Position	Professor of Psychological Medicine

12. Acknowledgement, Review and Submit

Conflict checks: Please declare any conflicts or potential conflicts of interest that you may have in undertaking this research, including any relevant, non-personal & commercial interest that could be perceived as a conflict of interest.

By completing this form, the programme reserves the right to share, in confidence, details of your application with other parts of the NIHR and with the Department of Health and Social Care.

Agreement to Terms and Conditions: I confirm that I have secured all necessary licences and ethical approvals in relation to the research and will abide by the terms of those licences and approvals in carrying out the research. I confirm that I have read the NIHR Carbon Reduction Guidelines and, where possible, have taken steps to reduce the carbon emissions generated by this research. I confirm that all material within this application, with the exception of sections that other participants are required to complete, is my own work and I have taken steps to ensure that no parts of the application have been inappropriately plagiarised. I undertake to conduct the research according to such guidelines for good research practice as the Host Organisation may from time to time lay down according to the NIHR Research Governance Framework. If you are applying for an NIHR award which comes with partnership funding from another organisation (e.g. a charity or industry partner) your information (including your NIHR Application Form and subsequent progress/annual reports) may be shared with the partner organisation as part of the review process and ongoing monitoring of the award. In completing this, I declare that, to the best of my knowledge, the information provided in this application, is true, accurate and complete. I have read the Guidance Notes and agree to accept the process by which an application is assessed and agree to abide by the conditions under which an award may be granted.
☒Confirmed

Appendices

1. NIHR CKD CVD 25 references.docx
2. Appendix.docx
3. NIHR PhD 25 Timetable - Gantt Chart -final.pdf
4. SoECAT Requirement Justification.pdf

Appendix #1: NIHR CKD CVD 25 references.docx

References

1. Hull SA, Williams C, Ashworth M, Carvalho C, Boomla K. Prevalence of suspected COVID-19 infection in patients from ethnic minority populations: a cross-sectional study in primary care. Br J Gen Pract. 2020 Oct 1;70(699):e696–704.
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Appendix #2: Appendix.docx

Appendix – Medication Classification Groups

Antihypertensives	Renin-Angiotensin System Inhibition	Lipid-Lowering Therapy	Sodium-Glucose Cotransporter-2 Inhibitors	Glucagon-Like Peptide 1 Agonists	Finerenone
Calcium Channel Blocker including: Amlodipine Nifedipine Felodipine	Angiotensin Converting Enzyme Inhibitor including: Captopril Enalapril Lisinopril Perindopril Ramipril	Statins including: Atorvastatin Fluvastatin Pravastatin Rosuvastatin Simvastatin	Canagliflozin	Liraglutide	
Diuretic including Thiazide and Thiazide-like including: Hydrochlorothiazide Indapamide Metolazone Furosemide Bumetanide	Angiotensin II Receptor Blockers including: Losartan Valsartan Irbesartan Olmesartan Telmisartan Candesartan	Ezetimibe	Dapagliflozin	Dulaglutide	
		Others including Fibrates, Bempedoic acid, inclisiran	Empagliflozin	Semaglutide	
			Ertugliflozin	Exenatide	
				Lixisenatide	

Appendix – KDIGO CKD Staging

CURRENT CHRONIC KIDNEY DISEASE (CKD) NOMENCLATURE USED BY KDIGO

CKD is defined as abnormalities of kidney structure or function, present for a minimum of 3 months, with implications for health. CKD is classified based on Cause, Glomerular filtration rate (GFR) category (G1–G5), and Albuminuria category (A1–A3), abbreviated as CGA.

KDIGO: Prognosis of CKD by GFR and albuminuria categories				Persistent albuminuria categories Description and range		
				A1	A2	A3
				Normal to mildly increased	Moderately increased	Severely increased
				<30 mg/g <3 mg/mmol	30–300 mg/g 3–30 mg/mmol	>300 mg/g >30 mg/mmol
GFR categories (ml/min/1.73 m²) Description and range	G1	Normal or high	≥90			
	G2	Mildly decreased	60–89			
	G3a	Mildly to moderately decreased	45–59			
	G3b	Moderately to severely decreased	30–44			
	G4	Severely decreased	15–29			
	G5	Kidney failure	<15			

Green: low risk (if no other markers of kidney disease, no CKD); Yellow: moderately increased risk; Orange: high risk; Red: very high risk. GFR, glomerular filtration rate.

Taken from : Stevens PE, Ahmed SB, Carrero JJ, Foster B, Francis A, Hall RK, et al. KDIGO 2024 Clinical Practice Guideline for the Evaluation and Management of Chronic Kidney Disease. Kidney International. 2024 Apr 1;105(4):S117–314.

Appendix #3: NIHR PhD 25 Timetable - Gantt Chart -final.pdf

	Objectives	2025			2026												2027												2028												2029					
#	Doctoral Specific Aims	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun
WP1:	Systematic review																																														
1.1	Protocol																																														
1.2	Data Sources, Selection Criteria & Search Strategy																																														
1.3	Quality Assessment and Data Synthesis																																														
1.4	Write up																																														
WP2:	Cohort Characterisation																																														
2.1	Data extraction																																														
2.2	Data Analysis																																														
2.3	Discussion and write-up																																														
WP3:	Decision Modelling Study																																														
3.1	Review of Decision model and input parameters																																														
3.2	Data wrangling																																														
3.3	Application of Decision Model																																														
3.4	Health Impact Modelling and Analysis																																														
3.5	Financial Impact Modelling and Analysis																																														
3.6	Scenario modelling																																														
3.7	Discussion and write-up																																														
4	Patient and Public Involvement																																														
5	Core PPI Group																																														
6	Collaborators andAdvisory Panel																																														
7	Courses																																														
7.1	NIHR Future-Focused Leadership programme																																														
7.2	Online Inequality in health and health care																																														
7.3	Systematic Reviews and Meta-analyses of Health Research																																														
7.4	Causal Inference in Epidemiology: Recent Methodological Developments																																														
7.5	Pharmacoepidemiology																																														
7.6	Statistical Methods in Economic Evaluation																																														
7.7	Online Courses in Decision Modelling for Health Economic Evaluation																																														
7.8	UK EQUATOR Centre's Publication School																																														
7.9	PPI Courses																																														
8	Full doctoral thesis Write-up																																														
9	Conferences & Presentations (optional)																																														
9.1	SAPC																																														
9.2	RCGP																																														
9.3	UK Kidney Week																																														
9.4	European Renal Association																																														
9.5	American Heart Association																																														
9.6	Local meetings and PPI Presentations (ad hoc)																																														
10	Post-doctoral Planning																																														
10.1	Apply for post-doctoral roles																																														

Appendix #4: SoECAT Requirement Justification.pdf

Statement Regarding SoECAT Form Requirement

A SoECAT form is not required for this study as it is based on secondary data analysis and does not involve NHS patients, staff, or resources for primary data collection.

This requirement has been previously discussed and confirmed with a member of the Study Support Service Officer, LCRN North Thames Core Team, NIHR Clinical Research Network (CRN).

This application was previously submitted last year and has been revised based on feedback. However, the core research methods of using secondary data remains the same. For reference, the SoECAT form submitted last year is attached to this application.

Schedule of Events Cost Attribution Template (SoECAT)
Study Information - template A

Version number 1.19 11 March 2020.
Please check the CRN website for the current version of this template.

Guidance
The guidance given on each tab should be read in association with the separate guidance document.

Please answer each question below (**NOTE: KEY FUNCTIONALITY NEEDED TO COMPLETE THE REST OF THIS TOOL IS NOT ACTIVATED UNTIL YOU HAVE PROVIDED AN ANSWER TO QUESTION 3**). Please only provide an IRAS reference number if you have already created an IRAS project for this study. All cells are free text other than questions 3, 5 and 11, which should be answered using the drop-downs, question 9, which auto populates from the dates given in answer to questions 7 and 8, and 10 which must be a whole number. Whilst we appreciate that it may be difficult to state with certainty answers to questions 7,8 and 10 best estimations should be provided to support the calculations within this tool. Questions 11, 12, 13 and 14 are for office use only.

1. IRAS Reference Number:	1009672
2. Short Study Title:	Real-World Comparative Effectiveness of Cardio-Renal Risk Reduction Interventions in CKD
3. Are you making application for funding from a portfolio eligible Association of Medical Research Charities (AMRC) member?	No
4. Funder Name	NIHR
5. Number of Study Arms	1
6. Chief Investigator Name	Dr Chris Carvalho
7. Planned Start Date	01 October 2024
8. Planned End Date	30 June 2028
9. Duration (Months)	45
10. Projected Number of Sites	1

OFFICE USE ONLY	
11. Lead Local Clinical Research Network (LCRN) or Devolved Administration	North Thames

12. Pre-application AcoRD Specialist Authorisation		Select yes for each that applies
This SoECAT has been completed for the named study-level application and the cost attribution is in line with the AcoRD guidance as of the given date		
SoECAT completion is not required because the study will not be included on the NIHR Clinical Research Network Portfolio, or equivalent in Scotland, Wales or Northern Ireland. The applicant has been advised that the study will not be eligible to have its NHS Support Costs or Excess Treatment Costs recompensed.		
SoECAT completion is not required because there are only Research Costs. The applicant has been advised that an IRAS schedule of events will need to be completed at IRAS Approval application stage, if the study is to take place in the NHS (including in HSC in Northern Ireland).		yes
The application is for a Programme Grant. The applicant has been advised that the SoECAT will need to be updated once trial protocols have been developed.		
Name of AcoRD Specialist	Harshinder Virdee	
Date Authorised	25/01/2024	
Additional clarifications from AcoRD Specialist		

13. Main Commissioner	

14. Post Award Triage Authorisation	
Name of Authoriser	